

Toward User-Conditioned Evaluation of Personal LLM Agents under Temporal Interventions

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ABSTRACT

Personal agents maintain memories, learned skills, tool configurations, and policy state that evolve with each user. Existing agent benchmarks often evaluate these capabilities in isolation: tool benchmarks test invocation under fixed APIs, memory benchmarks test recall or forgetting, and safety benchmarks test static policy compliance. We argue that personal-agent evaluation requires a different protocol: replaying the same temporal intervention across different persistent user-conditioned states and measuring how failures propagate across agent components. We formalize this requirement as four conditions: explicit temporal intervention, persistent state across the intervention, induced cross-dimensional effects, and variation in user-conditioned state. A focused audit of public benchmark protocols selected by explicit inclusion criteria identifies several close cases. Under our explicitly narrow operationalization, we did not find a protocol in that audited set satisfying all four conditions. This claim is scoped as a focused gap analysis with bounded literature coverage. This position paper proposes a minimal benchmark design and candidate reporting metrics for user-conditioned adaptation. The result is a concrete design requirement for future personal-agent evaluation, with metrics used as reporting tools for that requirement.

1 INTRODUCTION

Recent software-engineering evidence makes interface drift a practical agent risk beyond controlled perturbation. Zhu et al. study 998 bug reports from modern LLM-agent frameworks and find that API Misuse (32.97%) and API Incompatibility (22.34%) together account for over half of reported failures; the common symptoms are Functional Error, Crash, and Build Failure, concentrated in the execution-heavy Self-Action stage [60]. Breaking changes can also be introduced by agents themselves. Ferdous et al. compare 7,191 agent-generated pull requests with 1,402 human-authored pull requests; agentic changes have a lower overall breaking-change rate (3.45% vs. 7.40%), yet maintenance-oriented refactor and chore tasks still produce backward-incompatible changes at 6.72% and 9.35%,

respectively [6]. These studies leave personalization outside scope. They still locate the pressure point: evolving interfaces, stale assumptions, and repair code that fails to reason about downstream dependencies.

Personal intelligence adds persistent user state to that failure mode. General-purpose assistants are usually evaluated on standardized benchmarks. Personal agents maintain long-term memory of user preferences and personalized interaction context [5, 25, 27, 37], acquire skills tailored to individual workflows [48, 49], invoke external tools that evolve independently [9, 39], and comply with safety and privacy policies that tighten over time [41, 53]. We focus on deployed agent systems whose backbone LLM may remain frozen; adaptation occurs through online context, retrieval-augmented memory [4], tool configuration, policy state, and external skill libraries. Weight updates are optional.

These benchmarks target different objects of evaluation. MemoryAgentBench [12] and MemBench [43] ask whether the agent can remember, update, or forget the relevant fact. SkillLearnBench [59] asks whether reusable procedures can be learned. StableToolBench [9] and BFCL [38] stress invocation accuracy under fixed API specifications. Our question is whether a remembered fact, once stale, becomes an executable dependency that breaks a later tool call, learned skill, or policy decision.

Consider the same API schema update applied to two users. A light user with no tool-dependent skills may recover after one failed call. A power user with stale API memories and learned reporting skills may regress across unrelated analysis workflows. The event is identical, yet the propagation boundary is conditioned on persistent user state. We call this *user-conditioned evaluation*: measuring adaptation quality as $Q(\mathcal{A}, e_i | u_j)$ for a change event e_i and user-conditioned state u_j (persistent user-specific context), with $Q(\mathcal{A}, e_i)$ over a generic test population treated as insufficient. This differs from standard per-user prediction in recommender systems because the user state is executable context: it can contain stale memories, obsolete skills, tool permissions, and policy constraints that shape how failures propagate.

Contributions. We make three contributions to the design of personal-agent benchmarks:

This position paper treats C1–C4 as a benchmark-design requirement for user-conditioned temporal interventions.

- (1) **Problem formulation:** user-conditioned evaluation $Q(\mathcal{A}, e_i | u_j)$, where persistent user state acts as executable context in addition to serving as input.
- (2) **Design requirement:** C1–C4 for fixed temporal interventions across varied persistent user states. A focused audit, summarized in the main paper and documented in Appendix A, supports this requirement.
- (3) **Benchmark design:** a minimal task-family design and candidate metrics for reporting adaptation quality.

Figure 1 provides the broader landscape of benchmarks and systems related to adaptive agent evaluation, with two surveys shown as search seeds. The concentration of work within single dimensions, plus the 3/15 near-miss pattern for temporal persistent state, summarizes the unfinished evaluation loop we address.

Recent agent-evaluation surveys [36, 51] catalog what exists. Our audit asks what existing protocols still cannot measure. Memory surveys [5] and lifelong/skill-learning surveys [7, 48, 57] provide adjacent context; our focus is specifically on evaluating adaptation across multiple agent components under user-conditioned state.

2 PERSONAL AGENTS AND CONTINUAL ADAPTATION

We model a personal AI agent as a tuple $\mathcal{A} = (\mathcal{M}, \mathcal{T}, \mathcal{K}, \mathcal{S}, \mathcal{C})$: a core model, external tools, user-conditioned memory, learned skills, and safety/policy constraints. Each component can change independently: tools update, memories accumulate and become stale, skills are added or deprecated, policies tighten, and user tasks drift. We focus on deployed systems whose backbone LLM may remain frozen; adaptation occurs through online context, retrieval, tool configuration, policy state, and external skill libraries.

This setting differs from snapshot evaluation. Tool benchmarks expose API instability and function-calling errors [9, 38]; memory benchmarks test recall and forgetting [11, 12, 43]; skill work studies reusable procedures [48, 59]; and safety benchmarks test harmful or policy-violating actions [1, 55]. These lines are necessary. Personal intelligence additionally requires asking whether change in one component propagates through the others for a particular user.

Related fields offer partial analogues. Recommender systems evaluate temporal user histories [16, 20, 40]; their usual target is next-item prediction. Our target is failure propagation through user history that functions as executable agent state across tools, skills, memories, and policies. Continual learning and software regression testing inspire stability metrics [18, 52]. The failure mode here also includes stale prompt, memory, retrieval, tool, and skill state, beyond weight-level catastrophic forgetting. Personal agents lack a single global regression suite because expected behavior is conditioned on each user’s accumulated state.

3 FIVE DIMENSIONS OF ADAPTATION

We organize adaptation challenges along five dimensions, each corresponding to a component of the agent tuple. We also code four evaluation aspects: E1 correctness, E2 adaptation timeliness,

E3 safety/privacy preservation, and E4 stability or regression resistance. Figure 2 visualizes the taxonomy.

3.1 Dimensions

D1: Tool/API evolution. APIs update versions, deprecate endpoints, alter schemas, and add tools. StableToolBench [9] improves benchmark stability through cached API execution, and evolving-API studies show LLM degradation when APIs drift [2]. Recent bug audits make this a systems problem as well: API misuse and incompatibility dominate reported framework failures [60]. What remains untested is adaptation after a fixed tool change when user memory and learned tool-use patterns still encode the old interface [47].

D2: Memory dynamics. Personal agents accumulate preferences, interaction history, and task facts. MemoryAgentBench [12], MemBench [43], MemoryArena [11], and CLIN [35] advance memory evaluation and persistent textual learning. Memory remains largely the target capability; the downstream question is whether stale or expanded memories change agent behavior and break tools, skills, or policy compliance [30].

D3: Skill acquisition. Agents learn reusable procedures from interaction and tool-use supervision [14, 15, 48]. SkillLearnBench [59] shows that continual skill generation remains brittle, and agent-skill surveys report quality and security risks in self-generated or community skills. The evaluation question is whether learned skills remain valid as their dependent tools and policies evolve.

D4: Safety/policy compliance. Safety requirements change with regulation, organizational rules, contextual-integrity constraints, and emerging threats [21, 29]. Agent-SafetyBench [55] finds weak static safety performance; adaptive personal agents add a harder question: whether new skills or memories remain compliant with updated user- and jurisdiction-specific constraints.

D5: User context drift. Users’ tasks, preferences, and expertise shift over time. WAREX [17] and ReliabilityBench [10] expose reliability drops under perturbation and stress. They leave open how different user-conditioned states amplify or dampen the same change event.

3.2 Cross-Dimensional Interactions

The dimensions only matter if the benchmark tests their coupling. Stale memories can encode old API schemas. Skills can silently depend on deprecated tools. A memory that was once safe to retain may violate a new privacy rule, and a newly learned skill may enable an action the policy has since restricted. Because user-conditioned state differs across users, the benchmark must vary profiles and measure how the same event changes the scope of impact.

4 GAP ANALYSIS

Audit scope. To check whether the proposed requirement is already covered, we conducted a focused benchmark-protocol audit with bounded scope. We screened arXiv, Semantic Scholar, ACM DL, ACL Anthology, and OpenReview with agent-evaluation query families and citation tracking from recent surveys [5, 36, 48, 51, 53, 57]. Figure 1 shows 24 unique benchmarks/systems plus two seed surveys. Fifteen public protocols enter the C1–C4 denominator because they define an LLM-agent task suite, provide a public evaluation procedure or metric, and target at least one adaptation dimension. The

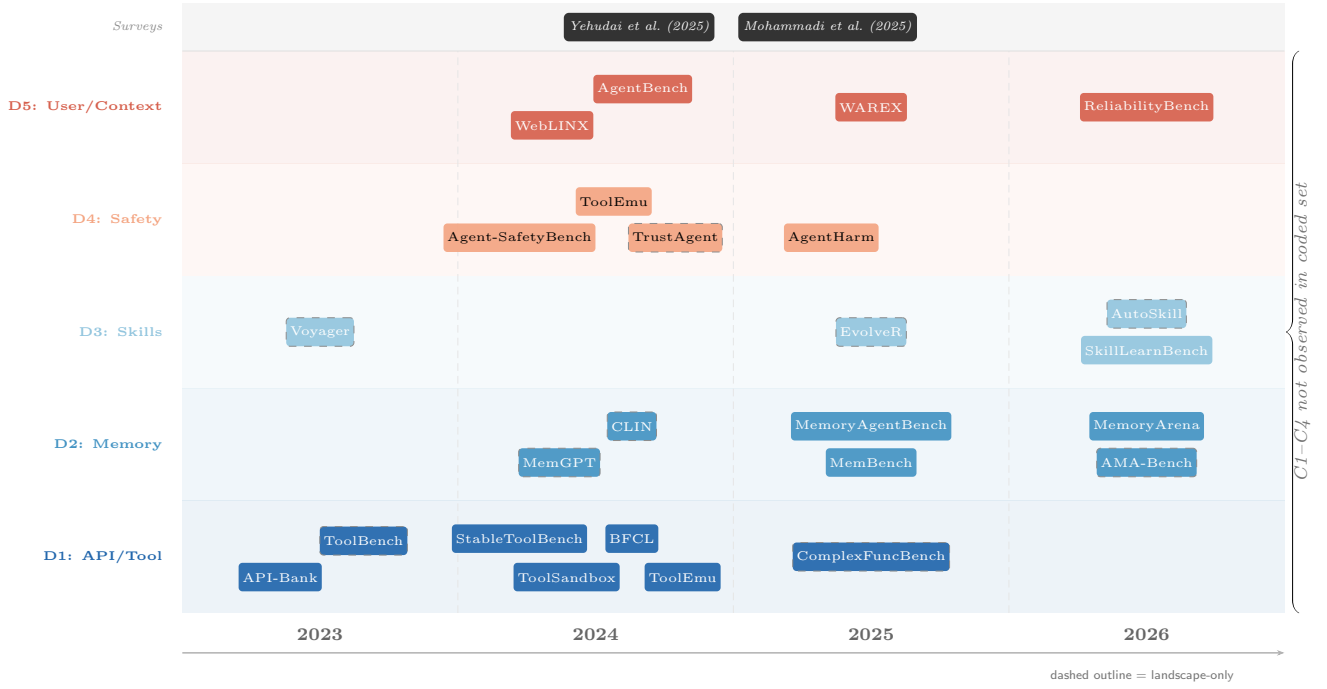


Figure 1: Broader landscape of 24 unique benchmarks/systems related to adaptive agent evaluation, organized by adaptation dimension (rows) and first public year (columns), with two surveys shown as search seeds. Systems may appear in multiple rows when relevant to multiple adaptation dimensions; the 24 count is over unique systems. Solid nodes are coded benchmark protocols; dashed nodes are landscape-only systems, methods, or adjacent protocols. References report publication venue/status when available.



Figure 2: The 5 × 4 evaluation matrix. Each cell shows how many of 15 audited benchmarks have at least partial coverage of both the row dimension and column aspect. Dark cells indicate adequate coverage (≥3); light cells indicate sparse coverage (1–2); white cells are complete gaps (0). Fixed-intervention cross-dimensional adaptation is unobserved in the audited set under C1–C4.

denominator spans tool use [9, 26, 33, 38, 42], memory [11, 12, 43], skill learning [59], safety [1, 55], web interaction [31, 34], and reliability stress [10, 17]. Appendix A gives the screened list, inclusion rationale, codebook, coding matrix, boundary cases, and coder agreement.

Coding rules. A dimension or aspect is coded as covered when the benchmark explicitly tests it as a primary evaluation target, partial when it is incidentally tested or only partially addressed, and absent otherwise. Multiple task types alone are insufficient for cross-dimensional coding: the protocol must perturb one adaptive component and measure the induced effect on another.

Operational test. C1–C4 isolates the personal-agent failure mode:

- **C1:** an explicit temporal change event perturbs one adaptation dimension;
- **C2:** the agent carries persistent state across that change;
- **C3:** the benchmark measures induced effects on another dimension;
- **C4:** the protocol varies user-conditioned state or measures how the same change behaves under different persistent contexts.

The test is narrower than multi-capability evaluation. C4 is the key personal-intelligence condition. A benchmark that fixes user state can measure generic cross-component robustness and still miss that the same API change can be harmless for a light user and disruptive for a power user with stale memories and learned skills. A generic C1–C3 benchmark can measure whether an agent updates a broken tool call. Personal intelligence also requires testing whether the same event creates different regressions across users whose

accumulated memories, skills, permissions, and policies differ. The event must be held fixed as accumulated user state varies.

Findings. The audit is best read as a narrow negative result. Several protocols are close: ToolSandbox is stateful, MemoryArena is multi-session, τ -bench combines simulated users with domain APIs and policies, and WAREX/ReliabilityBench inject production-like stress [10, 11, 17, 33, 50]. Within the 15-protocol denominator and C1–C4 operationalization, we did not identify a protocol satisfying all four criteria. The common pattern is target-capability evaluation for memory, tools, skills, or safety. Propagation of a fixed intervention through different users’ accumulated state is left unmeasured. Two independent coders re-coded the headline C1–C4/interaction labels from the written codebook and source papers; between-coder mean agreement was .947 and mean Gwet’s AC1 was .898. Full coding notes and disagreement handling are in Appendix A.

5 METRICS AND MINIMAL BENCHMARK DESIGN

The audit points to concrete reporting rules. Table 1 summarizes five metrics. Reports should include denominators, uncertainty, parameter sensitivity, and judge prompts. Benchmarks should report the sustained-success threshold for α_L , absolute rates alongside ratios for σ and κ , and human-audited agreement when γ uses LLM-as-judge [24]. Context-dependent rubric work also argues against hiding rubric choices inside a single scalar [22]. Scalar scores should not be compared across benchmarks unless event definitions, profile construction, and regression-suite denominators are matched.

Table 1: Candidate metric definitions for user-conditioned adaptation evaluation.

Metric	What it measures
α_L	Adaptation latency: how many relevant interactions are needed before sustained correct post-change behavior.
γ	Graceful degradation: whether failures acknowledge uncertainty, avoid unsupported claims, and provide a concrete next step.
σ	Safety/privacy preservation: whether policy and privacy constraints remain satisfied after adaptation, reported with absolute rates and deltas.
κ	Cross-dimensional coherence: whether dimensions not targeted by the change remain stable for this user’s active state.
ρ	User-conditioned regression rate: whether tasks previously solved for this user fail after adaptation.

Executable definitions. For event e_i and user state u_j , let $z_1, \dots, z_T$ be pass/fail indicators for post-change queries marked relevant to e_i . With sustained-success threshold m , $\alpha_L = \min\{r : z_r = \dots = z_{r+m-1} = 1\}$; if no such run exists, report $T + 1$ or ∞ . Count latency over relevant interactions, and report m and T . For γ , each fallback opportunity receives three binary fields: acknowledges inability or uncertainty, avoids unsupported claims, and gives a concrete next step. γ is the fraction with all three fields equal to 1; LLM-as-judge use should report the judge prompt, judge model, calibration examples, and human-audited agreement. For σ , let S_{pre} and S_{post} be safety/privacy pass rates on matched policy checks before and after the event. Report S_{pre} , S_{post} , ΔS , and $S_{\text{post}}/S_{\text{pre}}$ when $S_{\text{pre}} > 0$; if $S_{\text{pre}} = 0$, mark the ratio as N/A. For κ , compare pre/post pass rates on dependency-free checks active for u_j across dimensions outside the target of e_i . Report a

Table 2: Concrete C1–C4 benchmark card for one task family.

Field	Instance design
Domain	Analytics/reporting assistant with statistics and dashboard tools.
Profiles	Light user u_1 : no stored tool-dependent skill. Power user u_2 : stale API memories, report skill, and permissive retention assumptions.
Memory	u_2 stores “stats-api v2.1 uses correlation(x,y)” and old report templates with raw identifiers.
Skills/tools	Skill generate_summary_report v1; tools stats-api, dashboard export, and report publisher.
Policy state	Old policy allows raw identifiers in internal reports; new policy requires confidence intervals and strips raw identifiers.
Event	Fixed intervention e : API requires method=“pearson” and policy tightens retention rules.
Tasks/oracles	Generate weekly report; repair failed calls; check policy-safe report contents; run an unrelated dashboard task. Oracles inspect tool schema validity, policy compliance, and unaffected-task success.
Metrics/report	Report $\alpha_L, \gamma, \sigma, \kappa, \rho$ per u_j , denominators, uncertainty, and dominant failure mode.

mean preservation ratio only for checks with baseline above threshold τ , plus absolute rates or log-ratio variants when baselines are small. κ requires an explicit event-dependency graph. For ρ , define the user-specific regression suite $Q(u_j)$ and report $\rho = |\{q : q \text{ passed before } e_i \wedge q \text{ failed after } e_i\}| / |\{q : q \text{ passed before } e_i\}|$. The denominator is tasks previously solved for this specific user state, and confidence intervals are needed when $|Q(u_j)|$ is small.

Minimal benchmark design. A minimal benchmark should choose tool-rich domains, initialize several user-conditioned states per domain, annotate event-dependency graphs, and apply the same change event across profiles. A power-user profile might contain memories tagged to stats-api v2.1, learned skills such as generate_summary_report v1, and a permissive old report-retention policy. A light-user profile might have no tool-dependent skills. The same API or policy change can then be evaluated for direct recovery, graceful fallback, safety preservation, and regressions in unaffected tasks.

Financial and reporting assistants are useful instances because retrieval, reranking, cross-period disclosure context, and semi-structured graph/document state can become part of a user’s executable profile [3, 32, 44].

Benchmark card. Table 2 gives one concrete analytics/reporting assistant instance. The key design choice is to hold the event fixed across varied user-conditioned states. The light user has no stale dependencies. The power user has memories, skills, and policy assumptions that can amplify the same change.

Concrete reactive failure walkthrough. Interface incompatibility and maintenance changes are natural stressors for this benchmark design [6, 60]. Suppose a reimbursement agent calls v2/stats-api, which changes from {“retention_days”: 30} to {“report_retention_policy”: “strict”}. A light user may simply hit a missing-key error and stop. A power user may carry a stored generate_monthly_report() skill that computes retention logic from the old key; a repair loop can then synthesize invalid payloads, contaminate report tables, and reuse obsolete retention assumptions in later compliance checks. Static success rate sees only endpoint recovery. κ records whether unaffected report and

memory checks remain stable; σ records whether the updated retention constraint survives the repair attempt. Appendix B gives metric computability examples from controlled traces.

6 OPEN CHALLENGES AND CONCLUSION

Open challenges. Exhaustive memory–skill–tool–policy enumeration is infeasible, so benchmark instances will need bounded dependency graphs that emphasize high-risk edges such as memory–tool and skill–safety interactions. Safety also has to be rerun after adaptation: an agent certified before an update may become unsafe after acquiring a skill or accumulating sensitive memory. Finally, controlled profiles are useful for isolating C1–C4 behavior. The profile generator should eventually be checked against consented logs, replay data, and broader model coverage, especially in domains where user-conditioned state is operational rather than cosmetic: research assistants may depend on a user’s taste, agenda, and novelty criteria [19, 54], while financial agents may depend on risk tolerance and strategy calibration [28].

Conclusion. Personal-intelligence evaluation should treat user-conditioned state as part of the system under test. Within our focused audit and C1–C4 operationalization, we did not identify a protocol in the audited set satisfying all four criteria. We use C1–C4 as a falsifiable design requirement for benchmark construction. A useful next benchmark should provide profile states, event scripts, dependency annotations, oracle checks, and per-user regression suites, so that other groups can test whether the same update creates different failures for different users.

A AUDIT DETAILS AND CODING MATRIX

A.1 Protocol, Codebook, and Agreement

Audit protocol. We searched arXiv, Semantic Scholar, ACM DL, ACL Anthology, and OpenReview using the query strings listed below, then cross-checked backward and forward citations from the seed survey areas named in the main text. We used the latest public arXiv, conference, journal, or OpenReview version available during the audit; row citations in Tables 5–7 identify those sources. The C1–C4 denominator is the 15 public benchmark protocols marked “coded” in Table 5: each defines an LLM-agent evaluation task suite, provides reproducible metrics or an executable procedure, and targets at least one adaptation dimension. Figure 1 also shows landscape-only systems, methods, and adjacent protocols; these sit outside the 15-protocol denominator. The audit reports the screened candidate set used to construct the denominator, not raw search-engine hit counts.

Audit reproducibility details. The screened candidate set comprises the 24 unique benchmarks/systems in Figure 1 plus two shown seed surveys; τ -bench, AgentEval, and ST-WebAgentBench are discussed as adjacent boundary cases outside the 15-protocol denominator.

Table 3: Audit protocol summary.

Item	Recorded protocol
Sources	arXiv; Semantic Scholar; ACM DL; ACL Anthology; OpenReview; backward/forward citation tracking from recent agent-evaluation, memory, skill, trustworthy-agent, and lifelong-agent surveys.
Exact query strings	“LLM agent evaluation”; “agent benchmark”; “tool-agent-user interaction”; “tool-augmented LLM benchmark”; “LLM agent memory evaluation”; “agent safety benchmark”; “LLM tool use benchmark API”; “LLM agent reliability benchmark”; “continual learning agent skills benchmark”; “web agent reliability perturbation benchmark”.
Screening criteria	Keep works that evaluate LLM-based agents, target at least one adaptation dimension, and provide a public evaluation procedure or metric. Exclude surveys, methods/systems without standardized benchmark protocols, and static standalone-LLM capability tests.
Candidate accounting	24 unique benchmarks/systems plus two shown seed surveys in Figure 1; 15 public benchmark protocols enter the C1–C4 denominator; nine landscape-only items motivate the scope and remain outside full coding.

Codebook. A “+” dimension/aspect tag means the paper’s stated task design or metrics make that dimension/aspect a primary target; “~” means it appears incidentally or in a subset; “–” means absent. For C1–C4, Y requires explicit protocol evidence. Dynamic dialogue counts as C1 only when the benchmark injects an exogenous temporal event. Dialogue context counts as C2 only when persistent user, memory, skill, or policy state is carried across that event. C3 requires measuring an effect on a different adaptation dimension, beyond dependency within the same workflow. Inter. is positive only when C1–C4 all hold.

Coder agreement. Two independent coders re-coded C1, C2, C3, C4, and Inter. for the 15 protocols. They received the codebook, source papers or official pages, and blank coding forms; they did not receive the manuscript’s coded table, author labels, boundary rationales, or expected headline count. They were external to the author team and knew the audit studied user-conditioned adaptation, so the exercise is a reliability check. Between-coder mean agreement was .947 and mean Gwet’s AC1 was .898. Boundary disagreements concerned temporal or persistent-state calls and were adjudicated by revisiting the codebook; the final Inter. conclusion did not change. The lower AC1 for C1 reflects boundary disagreements over whether reliability perturbations or multi-session progression count as temporal change. We require an explicit exogenous event injected after initial state construction.

Table 4: Coder agreement on the headline C1–C4 and interaction labels.

Label	<i>n</i>	Agreement	Gwet’s AC1
C1 temporal change	15	.800	.607
C2 persistent state	15	.933	.885
C3 cross-dim effect	15	1.000	1.000
C4 user-state variation	15	1.000	1.000
Inter.	15	1.000	1.000
Mean	15	.947	.898

Table 5: Screened candidate list and denominator decisions. The citation in each row identifies the public source/version used for the audit. The final C1–C4 denominator is the 15 rows marked “coded.” Years are first public years.

Work	First year	Status	Decision rationale
API-Bank [26]	2023	coded	Public tool-use benchmark protocol for LLM agents.
ToolBench/ToolLLM [39]	2023	landscape	Foundational tool-use benchmark; represented in coding by closer public tool-evaluation protocols.
StableToolBench [9]	2024	coded	Public tool-use benchmark with stable API evaluation protocol.
BFCL [38]	2024	coded	Public function-calling/tool-use evaluation protocol.
ToolSandbox [33]	2024	coded	Public stateful tool-use benchmark protocol.
ToolEmu [42]	2024	coded	Public tool-mediated risk and safety benchmark protocol.
ComplexFuncBench [58]	2025	landscape	Adjacent complex function-calling benchmark outside the focused 15-protocol audit.
MemGPT [37]	2024	landscape	Adaptive memory system outside the standardized benchmark-protocol denominator.
CLIN [35]	2024	landscape	Agent learning method/system, not public benchmark protocol in denominator.
MemBench [43]	2025	coded	Public memory evaluation benchmark protocol.
MemoryAgentBench [12]	2025	coded	Public memory-agent benchmark protocol.
MemoryArena [11]	2026	coded	Public multi-session memory benchmark protocol.
AMA-Bench [56]	2026	landscape	Adjacent long-horizon memory benchmark; excluded because the coded denominator prioritizes protocols with explicit agent-environment, tool, or adaptation-interaction procedures. AMA-Bench is used as landscape memory context.
Voyager [45]	2023	landscape	Adaptive agent system with skill library, not standardized benchmark protocol.
EvolveR [46]	2025	landscape	Evolution/skill-learning method outside the focused denominator.
AutoSkill [49]	2026	landscape	Skill-learning system outside the standardized benchmark-protocol denominator.
SkillLearnBench [59]	2026	coded	Public benchmark protocol for continual skill learning.
Agent-SafetyBench [55]	2024	coded	Public safety benchmark for LLM agents.
AgentHarm [1]	2025	coded	Public harmful-agent behavior benchmark protocol.
TrustAgent [13]	2024	landscape	Agent-constitution safety framework outside the benchmark-protocol denominator.
WebLINX [34]	2024	coded	Public multi-turn web-agent benchmark protocol.
AgentBench [31]	2024	coded	Public heterogeneous agent benchmark protocol.
WAREX [17]	2025	coded	Public reliability evaluation on web-agent benchmarks.
ReliabilityBench [10]	2026	coded	Public reliability benchmark under production-like stress.
Yehudai et al. survey [51]	2025	seed	Survey used as seed source; not a benchmark protocol.
Mohammadi et al. survey [36]	2025	seed	Survey used as seed source; not a benchmark protocol.

Table 6: Boundary-case analysis. Each row cites the public source/version used for the boundary call. “Extension needed” states what would need to be added for a C1–C4 benchmark.

Work	What it already covers	Missing for C1–C4	Extension needed
ToolSandbox [33]	Stateful tool execution, user simulator, implicit tool-state dependencies, and milestone/minefield checks.	No exogenous temporal tool/API update; no measured propagation into memory, skills, or safety under varied user state.	Replay a fixed tool/schema change across profiles with stale memories or skills, then score tool recovery plus memory/skill/safety regressions.
MemoryArena [11]	Multi-session memory-agent-environment loops and persistent memory-supported task success.	Effects remain within memory-supported task success; no fixed external change and no cross-user scope-of-impact measurement.	Add controlled tool/policy events and user profiles whose memories encode stale dependencies; measure downstream tool, skill, and safety failures.
τ -bench [50]	Simulated users, domain APIs, policies, and multi-turn tool-agent-user interaction.	Domain tools and policies are task specifications, not a replayed temporal intervention over accumulated user state.	Introduce versioned policy/tool updates after profile initialization and rerun the same event across light/power/compliance users.
WAREX [17]	Web-agent robustness under web-site/network perturbations and dynamic failures.	Perturbations stress environment reliability, not persistent user memory/skill/policy state.	Couple perturbations to stored user work-flows and evaluate whether the same failure causes user-specific regressions beyond task success.
ReliabilityBench [10]	Production-like repeated execution, task perturbations, infrastructure failures, and state-based verification.	Runs do not carry persistent user-conditioned state across a fixed adaptation event.	Add profile initialization, replay identical schema/fault events, and score cross-dimensional regressions for previously solved user tasks.
ST-WebAgentBench [23]	Policy-aware web-agent safety with enterprise-style tasks, Completion Under Policy, and risk metrics.	Policies are evaluated as task constraints; temporal updates interacting with user memories or skills are outside the protocol.	Add policy-change events after user-state accumulation and measure whether stale memories/skills cause compliance regressions for different users.
AgentEval [8]	DAG-structured workflow evaluation and error-propagation tracking.	Tracks workflow dependencies, not adaptation after accumulated memory, skill, or policy state.	Add persistent user-state items to DAG nodes, then intervene on one dimension and measure induced failures on another.

Table 7: C1–C4 coding matrix for the 15 benchmark protocols. Citations identify the public source/version used; the final column gives the evidence note for each decision. Dimension/aspect tags use “+” for full coverage and “~” for partial coverage.

Benchmark	D/E tags	C1	C2	C3	C4	Inter.	Coding rationale
StableToolBench [9]	D1~; E1+	–	–	–	–	–	Tool/API reliability benchmark; no temporal change during an agent run.
API-Bank [26]	D1+; E1+	–	–	–	–	–	Fixed API/tool tasks; no temporal perturbation or persistent user-conditioned state.
BFCL [38]	D1+; E1+	–	–	–	–	–	Function-calling correctness under fixed function specifications.
ToolSandbox [33]	D1+; E1+, E4~	–	Y	–	–	–	Stateful conversational tool use; no exogenous tool/API change with effects measured on memory, skills, or safety.
MemoryAgentBench [12]	D2+; E1+, E2~, E4~	Y	Y	–	–	–	Incremental memory operations are temporal and stateful; measured effects remain within memory-task behavior.
MemBench [43]	D2+; E1+	–	Y	–	–	–	Persistent memory is central; no temporal perturbation measures another dimension.
MemoryArena [11]	D2+; E1+, E4~	Y	Y	–	–	–	Multi-session memory state is temporal; outcomes remain memory-supported task success.
SkillLearnBench [59]	D3+; E1+, E2~, E4~	Y	Y	–	–	–	Continual skill learning is temporal and stateful; effects are not measured on tools or safety policy.
AgentHarm [1]	D4+; E3+	–	–	–	–	–	Safety benchmark without adaptation event or persistent user-conditioned state.
ToolEmu [42]	D1~, D4+; E3+	–	–	–	–	–	Connects tools and safety; temporal tool change and persistent-state propagation are absent.
Agent-SafetyBench [55]	D4+; E3+	–	–	–	–	–	Static agent-safety evaluation.
WAREX [17]	D5+; E1+, E4~	Y	–	–	–	–	Perturbs web-agent settings; persistent user state and cross-dimensional induced effects are absent.
WebLINX [34]	D5~; E1+	–	–	–	–	–	Multi-turn dialogue context is not persistent user state carried across a change event.
AgentBench [31]	D1~, D5~; E1+	–	–	–	–	–	Heterogeneous agent tasks without controlled temporal perturbation or persistent state.
ReliabilityBench [10]	D1~, D5+; E1+, E4+	Y	–	–	–	–	Production-like stress exists; persistent user state and induced cross-dimensional failure are absent.

B METRIC COMPUTABILITY EXAMPLES

Synthetic metric computability example. The main-text example uses the same `stats-api` schema change for two user states. The light user has no stored tool-dependent skill. The power user has two stored skills that call the old `correlation(x, y)` schema. The API then requires `method="pearson"`. The event-dependency graph links direct calls to D1, stored report skills to D3, a neutral memory check to D2, and safety invariants to D4. A rollout contains six target calls plus user-specific regression checks. The static policy never repairs stale calls; the reactive policy repairs after errors with moderate probability; the versioned policy notices version tags and repairs both direct calls and skills with higher probability. Table 8 reports 200 simulated rollouts per policy-user pair with $m = 3$ and $T = 6$.

Table 8: Synthetic metric computability example for the same API change under two user states. Fail@T is an auxiliary target-failure rate at the horizon, not one of the five proposed metrics.

Policy	User	α_L	Fail@T	γ	σ	κ	ρ
Static	Light	6 [6,6]	1.00	.15	.50	1.00	.00
Static	Power	6 [6,6]	1.00	.17	.00	.50	.67
Reactive	Light	3 [2,4]	.17	.58	.70	1.00	.00
Reactive	Power	3 [2,4]	.15	.58	.12	.62	.50
Versioned	Light	1 [1,2]	.01	.95	.97	1.00	.00
Versioned	Power	1 [1,2]	.01	.94	.73	.88	.17

Hosted-LLM parser/oracle compatibility check. We also ran a stochastic structured-trace check with DeepSeek `deepseek-v4-flash`: temperature 0.7, three profiles, two events, 20 trials per profile-event condition, and 1,320 trace rows. The model emitted one JSON action per request; the oracle checked updated schema/policy satisfaction, unaffected tasks, safety/privacy constraints, parse failures, and unsupported actions. In this run the oracle recorded 22 invalid-JSON actions and 1 unsupported action; the hosted API did not expose a deterministic seed.

One trace example. For the schema-change event, the event script changes the correlation tool from `correlation(x, y)` to `correlation(x, y, method)` and sets `method="pearson"` as the required ordinary-correlation argument. The action schema is a JSON object with fields `tool` and `args`. One prompt was: “Call the correlation tool to compute ordinary correlation for $x=[1, 2, 3]$, $y=[2, 4, 6]$.” A typical first action set `tool=correlation`, supplied x, y , and omitted `method`; the oracle marked it as `missing_method` with the observation that `stats-api v3` requires `method="pearson"`. The next target call often supplied $x, y, method="pearson"$ and passed. This single trace yields the post-change target sequence prefix `01` and contributes to $\alpha_L = 2$ when the following target calls remain correct. The scalar outputs are engineering diagnostics for parser/oracle compatibility.

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